Vehicle Model Gazebo Documentation - Team 24

MCTR1002: Autonomous Systems

Milestone 01: Simulation and Validation

1 Vehicle Model in Gazebo

The vehicle is an Ackermann-steering platform imported into Gazebo. It is configured to receive movement commands and provide state feedback for validation.

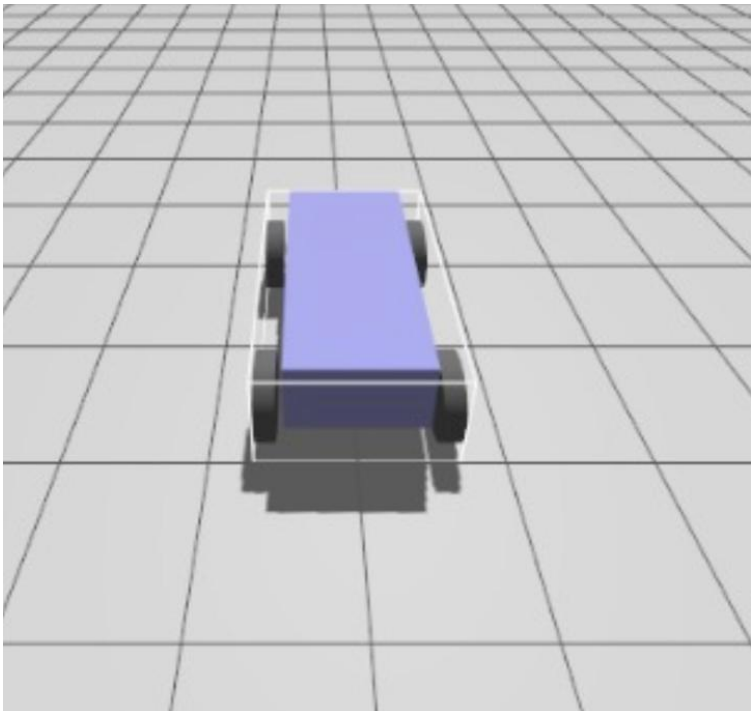

Figure 1: The Ackermann vehicle model within the Gazebo world.

2 Active ROS 2 Topics

The following topics are used to control the vehicle and monitor its status:

Topic Name	**Message Type**	**What it does?**
/model/vehicle_blue/cmd_vel	Twist	Receives speed and steering from keyboard.
/model/vehicle_blue/odometry	Odometry	Sends the car's (x, y) position to the node.
/tf	TFMessage	Handles the car's coordinates in the world.
/clock	Entity	Syncs simulation time between ROS and Gazebo.

3 Validation Node Functionality

The node `Validation_Printing_Node_Team_24.py` performs two main tasks:

1. **Health Check:** Prints "Hello World from Team 24" to prove the installation is correct.
2. **Position Feedback:** Subscribes to *odometry* to print the car's live (x, y) coordinates once per second.

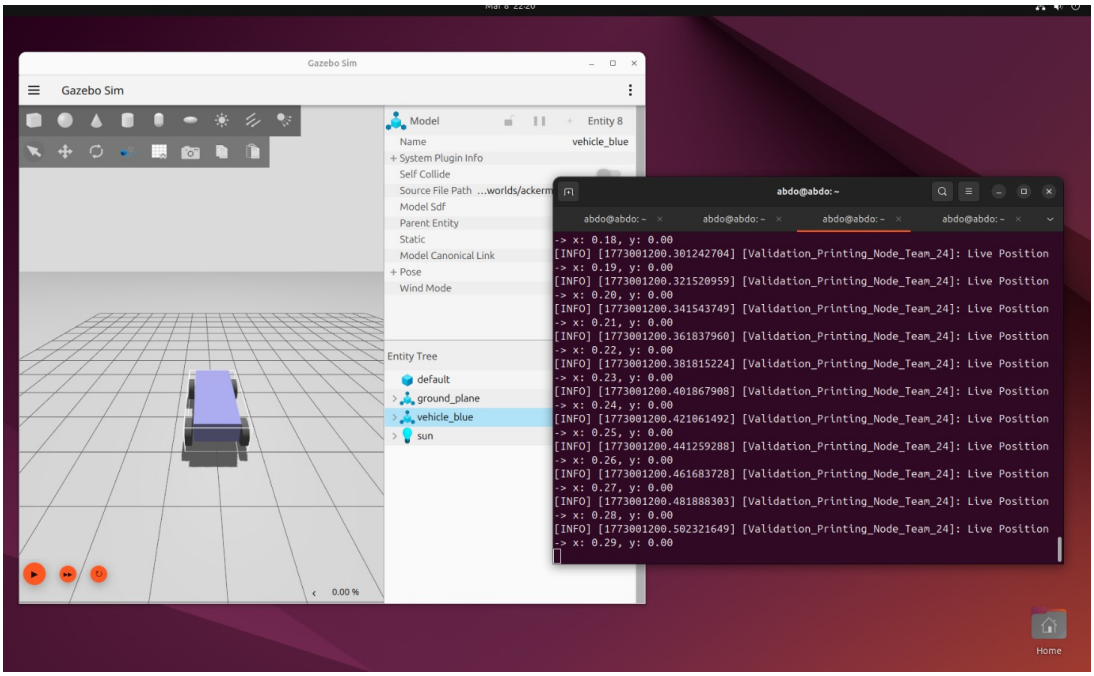

Figure 2: Terminal output showing the live position updates.

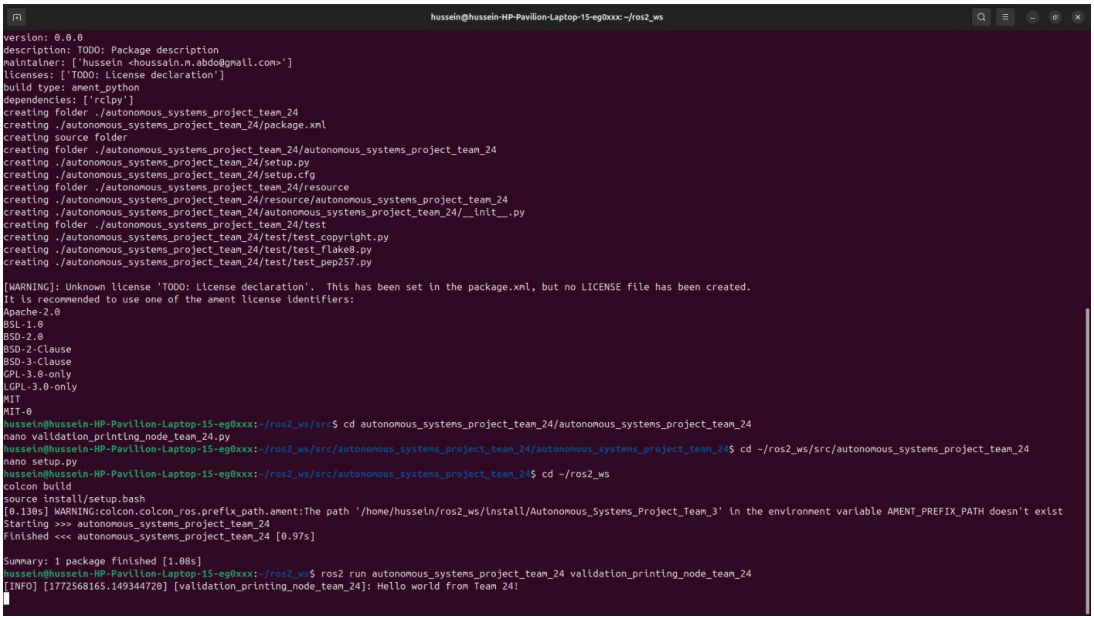

Figure 3: Terminal output showing the Hello World validation line.